Model scraping for comparison and validation

Summary			
Model name:	Cefep_1	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration route/mode *:	IV/3 min	No. compartments*:	3

Publication	
Title:	Pharmacokinetic-pharmacodynamic rationale for cefepime dosing regimens in intensive care units
Author(s):	Juliana F Roos , Jurgen Bulitta, Jeffrey Lipman, Carl M J Kirkpatrick
Other info:	PMID 16943209
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/16943209/

Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Critically ill patients with normal renal function (creatinine concentration of <0.1 mmol/L)		
Patient disease(s):	NA		
Drug dosing units*:	mg	Number of patients:	13
Plasma conc. units*:	mg/L	No. samples:	307
Daily dosing scheme	2000 q12h		
Notes/further info:			

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y	60				34-75
Weight	Kg	75				56-128
APACHE II	/71	11				4-24
Measured Creatinine clearance	L/h	7.1				2.3-11.7

Categorical covariate	Feature	Count (%)
Sex male	Yes	85
Notes/further info:		

Model scraping for comparison and validation

Final Model Structure			
Compartments:	3		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	no
BLLOQ handling method (mg/L):	1	Dose compartment:	central
All covariates tested:	Age weight serum creatinine creatinine clearance measured by 8 h urine collection creatinine clearance estimated via C&G equation using total body weight APACHE II		
Covariate inclusion method:	Possible covariates added in a stepwise fashion into the model.		
Covariates included*:	cefepime clearance : Creatinine clearance measured by 8 h urine collection (L/h)		
'Typical' patient for scaling:	$CLCR_{Std}$ 7 L/h		
Graphical representation / schematic	NA		
Equations	$TVCL = THETA_1^{*}(CLCR/CLCR_{Std})$ TVV1= THETA_2 TVV2=THETA_3 TVV3= THETA_4 TVQ2=THETA_5 TVQ3=THETA_6		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	6.51	5.22-8.01			
THETA_2 (L)	5.74	4.95-6.56			
THETA_3 (L)	9.57	7.86-11.7			
THETA_4 (L)	7.3	(fixed)			

THETA_5 (L/h)	33.8	29.4-38.4			
THETA_6 (L/h)	3.5	(fixed)			
Between-subject variability (inter-individual variability)					
OMEGA_ CL (CV%)		22-48	38		
OMEGA_V1		2-39	23		
OMEGA_V2		20-44	34		
Between-occasion variability (inter-occasion variability)					
KAPPA_CL (CV%)		7-22	15		
KAPPA_V1		21-35	30		
Residual error					
Proportional error	0.08				
Additive error (mg/L)	0.452				
Notes/further info:					

** required as a minimum for model replication*

Model scraping for comparison and validation

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	Yes, fig 1
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Summary			
Model name:	Cefep_2	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration route/ mode *:	IV/30 min	No. compartments*:	2
Publication			
Title:	A model-based analysis of the predictive performance of different renal function markers for cefepime clearance in the ICU		
Author(s):	Stijn Jonckheere, Nikolaas De Neve, Hans De Beenhouwer, Mario Berth, An Vermeulen, Jan Van Bocxlaer, Pieter Colin		
Other info:	PMID 27246236		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/27246236/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	NA		
Patient disease(s):	suspected or documented respiratory infection 85% catheter related bloodstream infection 5% neutropenic sepsis 5% infection of unknown origin 5%		
Drug dosing units*:	mg	Number of patients:	20
Plasma conc. units*:	mg/L	No. samples:	208
Daily dosing scheme	1000 post dialysis to 2000 q8h		
Notes/further info:	Dosing :		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y	75				69-78
Weight	Kg	76				67-90
BMI	Kg/m²	26.3				22.6-29.1
Body surface area	m²	1.9				1.8-2
SOFA score on ICU admission	/24	7				4.8-9.3
CG clearance	mL/min	49				31-123
Measured CL_{CR}	mL/min	47				26-106
Categorical covariate	**Feature**			**Count (%)**		
Sex male	Yes			75		

Intermittent haemodialysis	Yes	20
Mechanical ventilation	Yes	30
Notes/further info:	CG : Cockcroft-Gault	

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method:	NA	Dose compartment:	central
All covariates tested:	Cockcroft–Gault Measured CLCR MDRD CKD-EPI cystatin C Cystatin C in plasma CKD-EPI creatinine Creatinine in plasma Uromodulin in plasma Urea in plasmaKIM-1 in urine		
Covariate inclusion method:	P value was derived from the DOFV and the quantiles of a x2 distribution with 2 degrees of freedom and a value of ,0.05		
Covariates included*:	eGFR by Cockcroft–Gault formula (in L/h), DIAL (1 if dialysis, 0 otherwise, “For patients on IHD, we assumed that renal clearance was absent.”)		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	TVCLr= THETA_1*(1+ THETA_2×[(Cockcroft–Gault)–3.40])*(1-DIAL) TVV1=THETA_3 TVV2=THETA_4 TVQ2=THETA_5 TCLdialysis=THETA_6*DIAL TVCLother=THETA_7		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	2.88			13.8	
THETA_2 (h/L)	0.368			13.7	
THETA_3 (L)	18.3			13.7	
THETA_4 (L)	11,1			18.2	

THETA_5 (L/h)	6,63			60.9	
THETA_6 (L/h)	5,74			15.9	
THETA_7 (L/h)	0,87			10.1	
Between-subject variability (inter-individual variability)					
OMEGA_ CLr			49.5	48.9	
OMEGA_V1			39.7	58.7	
OMEGA_Q2			Not estimated	Not estimated	
OMEGA_V2			Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error (plasma)	0.332				
Proportional error (urine)	0.383				
Notes/further info:	CV (%) is calculated according to : √e omega−1×100%.				

Figures and diagnostic plots

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig S1, S2	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Summary			
Model name:	Cefep_3	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration route/mode *:	IV/30 min	No. compartments*:	2
Publication			
Title:	Population pharmacokinetic model of cefepime for critically ill adults: a comparative assessment of eGFR equations		
Author(s):	Erin F Barreto, Jack Chang, Andrew D Rule, Kristin C Mara, Laurie A Meade, Johar Paul, Paul J Jannetto, Arjun P Athreya, Marc H Scheetz		
Other info:	PMID 37882514		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC10648925/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	>36h cefepime Exclusion : acute kidney injury or on kidney replacement therapy or extracorporeal membrane oxygenation		
Patient disease(s):	Sepsis (82%)		

Drug dosing units*:	mg	Number of patients:	120
Plasma conc. units*:	mg/L	No. samples:	379
Daily dosing sheme	Majority of 2000 (90%) q12h (69%) regimen		
Notes/further info:			

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y		61	17		
weight	Kg		87	26		
BSA	m^2		2	0.3		
Serum creatinine	mg/dL		0.9	0.4		
Cystatin C	mg/L		1.3	0.6		
Charlson Comorbidity Index	/37	4				2-8
C-reactive protein	NA	77				31-144
APACHE III score	/299		88	31		
SOFA score	/24		7	4		

Categorical covariate	Feature	Count (%)
Sex male	Yes	60
>30kg/m^2	Yes	41
Comorbidies:		
Diabetes mellitus	Yes	25
Liver disease	Yes	4
Chronic kidney disease	Yes	19
Metastatic cancer	Yes	21
Transplant history	Yes	7
Corticosteroid exposure	Yes	43
Invasive mechanical ventilation	Yes	37
Vasopressor use	Yes	47
Sepsis	Yes	82
Dose 2g	Yes	90
Interval q12h	Yes	69
Notes/further info:		

Final Model Structure

Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	SAEM	Log-transformed?	Yes
BLLOQ handling method (mg/L)	0.5	Dose compartment:	central
All covariates tested:	Age sex race ethnicity presence of liver disease height body surface area body mass index weight eGFR (creatinine, cystatin C, or both based on the Cockcroft-Gault estimated creatinine clearance and the 2012 and 2021 CKD-EPI equations)		

Covariate inclusion method:	reduction in the BICc, −2 log likelihood (−2LL) and the rule of parsimony
Covariates included*:	CKD-EPI $eGFR_{cr\text{-}cysC}$ (mL/min), weight (kg)
'Typical' patient for scaling:	7.2 ml/min, 70kg
Graphical representation / schematic	NA
Equations	TVCL = THETA_1 * (eGFR_cr_cysc_2012 / 7.2) TVV1 = THETA_2 * (weight_td/70) TVQ=THETA_3 TVV2=THETA_4 *td: time dependant*
Notes/further info:	

Parameter Estimates* (adjust as appropriate)

PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	7.84			3.07	
THETA_2 (L)	15.56			9.31	
THETA_3 (L/h)	7.09			fixed	
THETA_4 (L)	10.63			fixed	
Between-subject variability (inter-individual variability)					
OMEGA_ CL (standard deviation)	0.24		23.9	9.68	
OMEGA_ V1	0.41		43.2	15.8	
OMEGA_ Q	Not estimated		Not estimated		
OMEGA_ V2	Not estimated		Not estimated		
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.27		5.65	5.65	
Additive error (mg/L)	0.97		8.55	8.55	
Notes/further info:					

Figures and diagnostic plots

Model Evaluation Metrics

Model scraping for comparison and validation

Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig S4	Example plasma conc. profiles:	Yes, fig S2
Other goodness-of-fit plots:	Yes, fig1, fig S5	Simulated plasma conc. profiles:	NA
Notes/further info:			

Summary			
Model name:	Cefep_4	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration route/mode*:	IV/30 min	No. compartments*:	2
Publication			
Title:	Population Pharmacokinetics and Monte Carlo Simulation of Cefepime in Critically Ill Patients with Hospital-Acquired/Ventilator-Associated Pneumonia		
Author(s):	Hyeonji Seo, Yong Kyun Kim, Sunghoon Park, Hwan-Il Kim, Dong-Hwan Lee		
Other info:	PMID 36450287		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/36450287/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Hospital-Acquired/Ventilator-Associated Pneumonia		
Patient disease(s):	Pneumonia		
Drug dosing units*:	mg	Number of patients:	21
Plasma conc. units*:	mg/L	No. samples:	144
Notes/further info:	2000 q8/12/24h		

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**Range**
Age	Y	67	67.10			
Height	Cm	165	164.05			
Weight	Kg	60	59.68			
Body surface area	m^2	1.60	1.64			
SOFA	/24	4	5.05			
APACHE II	/71	17	17.33			
Protein	g/dL	5.5	5.42			
Albumin	g/dL	2.8	2.87			
Blood urea nitrogen	mg/dL	22.1	22.48			
Scr	mg/dL	0.80	1.07			
CLCr, Cockcroft-Gault	mL/min	77.2	79.02			
Categorical covariate	**Feature**			**Count (%)**		
Sex male	Yes			57		

Model scraping for comparison and validation

Notes/further info:	Scre :serum creatinine		
Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.5	Dose compartment:	central
All covariates tested:	Sex age weight height body surface area (BSA) serum protein level serum albumin level serum creatinine level renal function: Cockcroft-Gault (CG) formulations, Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI), modified CKD-EPI, Modification of Diet in Renal Disease (MDRD), modified MDRD		
Covariate inclusion method:	P <0.01 (ΔOFV <-6.635 for one degree of freedom		
Covariates included*:	Creatinine clearance calculated using the CG formulation		
'Typical' patient for scaling:	77.21 mL/min		
Graphical representation / schematic	NA		
Equations	TVCL = THETA_1 * (CG/77.21) ^{THETA_2} TVV1=THETA_3 TVQ=THETA_4 TVV2=THETA_5		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	6.6	5.55-7.64		7.91	
THETA_2 (power)	0.656	0.438-0.796		10.7	
THETA_3 (L)	13.3	10.8-16.5		9.79	
THETA_4 (L/h)	16.5	9.59-23.7		18.7	
THETA_5 (L)	13	9.87-16.3		10.6	
Between-subject variability (inter-individual variability)					
OMEGA_ CL(CV%)		17-46.8	33.7	25.2	
OMEGA_ V1		7.52-49.5	34.1	24.9	
OMEGA_Q		0-65.8	50.8	18.4	
OMEGA_V2		16.7-54.3	40.1	21.2	
Between-occasion variability (inter-occasion variability)					
NA					

Residual error					
Proportional error	0.0762			9.43	
Notes/further info:					

Figures and diagnostic plots

Paste diagnostic plots etc from original publication here and compare to those generated in validation

---

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig S1	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Summary			
Model name:	Cefep_5	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration mode *:	IV/30 min	No. compartments*:	2
Publication			
Title:	Cefepime Population Pharmacokinetics, Antibacterial Target Attainment, and Estimated Probability of Neurotoxicity in Critically Ill Patients		
Author(s):	Muhammad Bilal, Michael Zoller, Uwe Fuhr, Ulrich Jaehde, Sami Ullah, Uwe Liebchen, Sören Büsker, Johannes Zander, Baharak Babouee Flury, Max Taubert		
Other info:	PMID 37366614		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC10353438/		
Source Study & Dataset			
Source study:	NCT01793012		
Inclusion criteria:	NA		
Patient disease(s):	Sepsis (except one)		
Drug dosing units*:	mg	Number of patients:	14
Plasma conc. units*:	mg/L	No. samples:	344
Daily dosing scheme	2000 q8/12/24h median daily dose 2000 q8h		
Notes/further info:			

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	Range
eCLCR	mL/min	60.8				16.7–217
mCLCR	mL/min	56				0–219
Body WT	Kg	70				45–120
Age	y	62				22–94
Body ht	m	1.68				1.5–1.83
BMI	Kg/m²	25				20–37
Serum albumin concn	mg/dL	2.5				1.8–3.6

Serum creatinine concn	mg/dL	0.8				0.3–4.4
Serum urea concn	mg/dL	53				12–196

Categorical covariate	**Feature**	
Sex male	Yes	43
Dialysis	Yes	29
Sepsis	Yes	93

Notes/further info:	eCLCR, Cockcroft-Gault creatinine clearance; mCLCR, measured creatinine clearance based on 24-h urinary excretion

Final Model Structure

Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method:	NA	Dose compartment:	
All covariates tested:	mCLCR eCLCR (not weight adjusted) total body weight age sex		
Covariate inclusion method:	change in OVF, with a change of 3.84, *P* value < 0.05 one degree of freedom		
Covariates included*:	creatinine clearance estimated by the Cockcroft-Gault equation (eClCR)), DIAL (1 if dialysis, 0 otherwise)		
'Typical' patient for scaling:	60.8 ml/min		
Graphical representation / schematic	NA		
Equations formula	TV CLND = (1-DIAL)*THETA_1* (eClCR/60.8) ^{THETA_3} TVCLD =DIAL*THETA_2 TVV1=THETA_4 TVQ=THETA_5 TVV2=THETA_6		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)

PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	2.94			10	
THETA_2 (L/h)	2.74			13	
THETA_3 (power)	0.896			17	
THETA_4 (L)	8.88			38	
THETA_5 (L/h)	27.4			54	

THETA_6 (L)	18.7			22	
Between-subject variability (inter-individual variability)					
OMEGA_CLD (CV%)			22.7	15	51
OMEGA_CLND			31.3	20	21
OMEGA_V1			47.3	25	14
OMEGA_V2			19.3	26	34
OMEGA_Q	Not estimated			Not estimated	
Between-occasion variability (inter-occasion variability)					
KAPPA_CLND	22.4			23	8
Residual error					
Proportional error	0.278			11	
Notes/further info:	CLD dialysis, ND non dialysis				

Figures and diagnostic plots

Paste diagnostic plots etc from original publication here and compare to those generated in validation

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig2	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	NA
Notes/further info:			

Summary			
Model name:	Cefep_6	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration route/mode *:	30 min	No. compartments*:	1
Publication			
Title:	Population pharmacokinetics and dosing simulations of cefepime in septic shock patients receiving continuous renal replacement therapy		
Author(s):	Mieke Carlier, Fabio S Taccone, Majorie Beumier, Lucie Seyler, Frédéric Cotton, Frédérique Jacobs, Jason A Roberts		
Other info:	PMID 26208469		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/26208469/		
Source Study & Dataset			
Source study:			
Inclusion criteria:	Severe sepsis/shock + AKI with CRRT Exclusion criteria: pregnancy, burns, cystic fibrosis Second study : additional inclusion criteria : CLcr<30 mL/min + at least one TDM sample, additional exclusion criteria : ECMO		
Patient disease(s):			
Drug dosing units*:	mg	Number of patients:	13
Plasma conc. units*:	mg/L	No. samples:	62
Daily dosing scheme	2000 q8/12h		
Notes/further info:	Data pooled from two studies		

Model scraping for comparison and validation

Patient Characteristics (adjust as appropriate)						
Continuous covariate	**Units**	**Median**	**Mean**	**SD**	**Mode**	**IQR**
Age	Y	59				43-70
Weight	Kg	70				65-75
SOFA score at the start of study	/24	10				9-14
APACHE II score on admission	/71	20				14-21
Blood flow rate	mL/min	150				140-150
Ultrafiltration flow rate	mL/h	1750				1500-2000
CRRT intensity	mL/kg/h	36				25-46
BMI	Kg/m²	24.5				23.9-27.2
Categorical covariate	**Feature**			**Count (%)**		
Sex male	Yes			54		
Anticoagulation	Heparin			69		
Notes/further info:						

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCE	Log-transformed?	Yes
BLLOQ handling method:	NA	Dose compartment:	Central
All covariates tested:	Age Sex Weight body mass index (BMI) Sequential Organ Failure Assessment (SOFA) score CRRT settings (blood flow rate, UFR (ultrafiltration rate), dialysis flow rate and CRRT intensity (= dialysate flow rate (mL/kg/h) + UFR (mL/kg/h)		
Covariate inclusion method:	decrease in OFV of 3.84 unit (P<0.05), comparative improvement in fit plot, biological plausibility		
Covariates included*:	UFR		
'Typical' patient for scaling:	UFR 1750 mL/h		
Graphical representation / schematic	NA		
Equations	TVCL = THETA_1× (UFR/1750) TVV1 = THETA_2 × (UFR/1750)		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	4.5	3.6-5.6			

THETA_2 (L)	40.8	33.4-48.7			
Between-subject variability (inter-individual variability)					
OMEGA_ CL (%CV)		17.1-49.5	37.7		
OMEGA_ V1		0.2-26.8	21.2		
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.204				
Additive error (mg/L)	3.3				
Notes/further info:					

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig1	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 2
Notes/further info:			

Summary			
Model name:	Cefep_7	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration route/mode *:	IV/30 min	No. compartments*:	1
Publication			
Title:	Adequacy of cefepime concentrations in the early phase of critical illness: a case for precision pharmacotherapy		
Author(s):	Erin F Barreto, Jack Chang, Matthew W Bjergum, Ognjen Gajic, Paul J Jannetto, Kristin C Mara, Laurie A Meade, Andrew D Rule, Kathryn J Vollmer, Marc H Scheetz		
Other info:	PMID 36648390		
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC10350476/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Critically ill adults treated with cefepime at an academic medical center between 2018 and 2020 exclusion criteria: exposure to cefepime within the 4-days prior to the study course greater than one dose of cefepime prior to ICU admission acute kidney injury stage II or higher kidney replacement therapy extracorporeal membrane oxygenation at the time of beta-lactam initiation		
Patient disease(s):	NA		

Drug dosing units*:	g	Number of patients:	100
Plasma conc. units*:	mg/L	No. samples:	220
Daily dose scheme	2000 (89%) q12h (70%)		
Notes/further info:			

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y		62	18		
Weight	kg		86	28		
BMI	Kg/m²		30	9		
Body surface area	m²		2	0.3		
Charlson comorbidity index	/37	4				2-9
C-reactive protein	NA	122				35-136
Body surface area	m²		2	0.3		
Cockcroft-Gault eCrCl			115	68		
CKD EPI $eGFR_{Scr}$			95	37		
CKD EPI $eGFR_{cysC}$			78	42		
CKD EPI $eGFR_{Scr-cyC}$			86	40		

Categorical covariate	Feature	Count (%)
Sex male	Yes	57
Sepsis	Yes	82
Pulmonary infectious site	Yes	63
Comorbidities:		
Diabetes mellitus	Yes	27
Liver disease	Yes	3
Chronic kidney disease	Yes	22
Metastatic cancer	Yes	31
Transplant history c	Yes	7
Corticosteroid exposure	Yes	40
Invasive mechanical ventilation	Yes	33
Vasopressor use	Yes	41
BMI>30	Yes	40
Notes/further info:		

Final Model Structure

Compartments:	1		
Elimination (e.g. 1st order):	1st order		
Estimation algorithm:	SAEM	Log-transformed?	Yes
BLLOQ handling method (mg/L):	0.5	Dose compartment:	central
All covariates tested:	Age Sex race/ethnicity weight body surface area (BSA) history of liver disease		

	eCrCl(CG), and estimated GFR based on creatinine (eGFRScr), cystatin C (eGFRCysC), or both (eGFRScr-CysC)
Covariate inclusion method:	BICc, goodness-of-fit checks of observed versus predicted concentrations, and between subject variability
Covariates included*:	log-normalized weight plus eGFRSCr-CysC on Cl, log-normalized weight on V
'Typical' patient for scaling:	NA
Graphical representation / schematic	NA
Equations	$TVCL = THETA1 * [(eGFRSCr\text{-}CysC/86)^{THETA_2} WT/86)]^{THETA_5}$ $TVV = THETA_3 * (WT/86)^{THETA_4}$
Notes/further info:	CG : Cockcroft-Gault

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_3 (L)	30.1			7.6	
THETA_4 (power)	1				
THETA_1 (L/h)	5.9			3.3	
THETA_2 (power)	0.81			6.7	
THETA_5 (power)	0.75				
Between-subject variability (inter-individual variability)					
OMEGA_V	0.21			36	
OMEGA_CL	0.21			13	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.24			9.3	
Additive error (mg/L)	1.47			8.3	
Notes/further info:	βV_{pop}: Coefficient for volume which represents a fixed allometric scaler for log(weight); βCL_{pop}: Coefficient for clearance which represents a fixed allometric scaler for log(weight) (see suppl mat.)				

Figures and diagnostic plots

Paste diagnostic plots etc from original publication here and compare to those generated in validation

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes fig S1	Simulated plasma conc. profiles:	Yes, fig 1

Notes/further info:	Figure S2. Distribution of individual parameters for volume of distribution and clearance in the final model

---

Summary			
Model name:	Cefep_8	Population type*:	ICU
Drug*:	cefepime	Model type:	POPPK
Administration route/mode *:	IV/0.08 to 1h (maj. on 0.5h)	No. compartments*:	1
Publication			
Title:	Population pharmacokinetics and target attainment analyses to identify a rational empirical dosing strategy for cefepime in critically ill patients		
Author(s):	Guohua An , C Buddy Creech, Nan Wu, Roger L Nation, Kenan Gu, Demet Nalbant, Natalia Jimenez-Truque, William Fissell, Pratish C Patel, Nicholas Fishbane, Amy Watanabe, Stephanie Rolsma, Carl M J Kirkpatrick, Cornelia B Landersdorfer, Patricia Winokur		
Other info:	PMID 37071586		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/37071586/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	NA		
Patient disease(s):	NA		
Drug dosing units*:	Mg	Number of patients:	130
Plasma conc. units*:	mg/L	No. samples:	278
Daily dosing scheme	Majority of 2000 q24h		
Notes/further info:	12 patients on RRT		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
Age	Y	65				52-72
Total Body Weight	Kg	90				70-110
Lean body weight	Kg	58				48-67
CLCR, LBW	mL/min	54				32-86
CLCR, TBW	mL/min	86				52-126
SOFA		6				4-9

Categorical covariate	Feature	Count (%)
Sex male	Yes	58
Sepsis	Yes	35
Septic shock	Yes	26
Mechanical ventilation	Yes	50
Acute kidney injury	Yes	35
Notes/further info:	LBW : lean body weight	

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st order		
Estimation algorithm:	FOCEI	Log-transformed?	Yes

Model scraping for comparison and validation

BLLOQ handling method (mg/L):	0.5	Dose compartment:	Central
All covariates tested:	Age sex TBW lean body weight, estimated creatinine clearance (calculated by the Cockcroft–Gault equation) based on TBW and lean body weight presence / absence of continuous renal replacement therapy (CRRT) mortality/severity of disease scores (SOFA) sepsis septic shock acute kidney injury abnormal hepatic function mechanical ventilation		
Covariate inclusion method:	Δ objective function value (OFV) >3.85; *P*<0.05] and backward elimination (ΔOFV >6.63; *P*<0.01		
Covariates included*:	creatinine clearance calculated according to **Cockcroft-gault** equation using lean body weight (CLCR,LBW) on cefepime CLR, CRRT, TBW		
'Typical' patient for scaling:	54 mL/min, 90 kg		
Graphical representation / schematic	NA		
Equations	TVCLT = (THETA_1 * (CLCR,LBW/54) * (1-FCRRT)) + (THETA_2 * FCRRT) + THETA_3 TVCLNR=THETA_3 TVV1= THETA_4 x (TBW/90) TVV2=THETA_5 TVQ=THETA_6 FCRRT = 1 for subjects on CRRT FCRRT = 0 for subjects not on CRRT		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)

PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	2			7.5	
THETA_2 (L/h)	1.64			14.8	
THETA_3 (L/h)	0.526			26.4	
THETA_4 (L)	13.4			17.2	
THETA_5 (L)	7.52			24.5	
THETA_6 (L/h)	12			35.2	
Between-subject variability (inter-individual variability)					

OMEGA_ CLT			29.9	22.3	
OMEGA_CLR			Not estimated	Not estimated	
OMEGA_CLNR			Not estimated	Not estimated	
OMEGA_V1			65.4	37.4	
OMEGA_V2			17.3	285.6	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.224				
Additive error (mg/L)	6.44				
Notes/further info:					

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Fig 3	Example plasma conc. profiles:	Fig 1
Other goodness-of-fit plots:	Fig S2	Simulated plasma conc. profiles:	Fig 2
Notes/further info:	Detailed history of model building and weight and renal function distribution in sup appendix		

Summary			
Model name:	Cefep_9	Population type*:	ICU
Drug*:	Cefepime	Model type:	POPPK
Administration route/mode*:	IV/30 min	No. compartments*:	2
Publication			
Title:	Population pharmacokinetics of four β-lactams in critically ill septic patients comedicated with amikacin		
Author(s):	Isabelle K. Delattre, Flora T. Musuamba , Philippe Jacqmin, Fabio S. Taccone, Pierre-François Laterre, Roger K. Verbeeck, Frédérique Jacobs , Pierre Wallemacq		
Other info:	PMID 22503878		
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/22503878/		
Source Study & Dataset			
Source study:	NA		
Inclusion criteria:	Patients with severe sepsis or septic shock treated with a loading dose of AMK (25 mg/kg) combined with meropenem		
Patient disease(s):	Sepsis or septic shock		
Drug dosing units*:	mg	Number of patients:	19
Plasma conc. units*:	mg/L	No. samples:	NA
Daily dosing scheme	2000 q8h		

Patient Characteristics (adjust as appropriate)

Continuous covariate	Units	Median	Mean	SD	Mode	IQR
NA						

Model scraping for comparison and validation

Categorical covariate	Feature	Count (%)
NA		
Notes/further info:	PKPOP model for meropenem, piperacillin, ceftazidime, cefepime	

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1st order):	1st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	NA	Dose compartment:	Central
All covariates tested:	Renal function		
Covariate inclusion method:	Empirical selection		
Covariates included*:	Creatinine clearance (Cockcroft–Gault) with a standardized WT = 70 kg		
'Typical' patient for scaling:	CLCrstd 100 mL/min		
Graphical representation / schematic	NA		
Equations	$TVCL = THETA_1 * (WT/70)^{THETA_2} * (CLCri/100)^{THETA_3}$ $TVV1=THETA_4 * (WT/70)$ $TVV2=THETA_5 * (WT/70)$ $TVQ=THETA_6 * (WT/70)^{0.75}$		
Notes/further info:	Model 1		

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h/70kg)	6.77			15.1	
THETA_2 (power)	0.75			NA	
THETA_3 (power)	0.41			40.3	
THETA_4 (L/70kg)	17.8			11.1	
THETA_5 (L/70kg)	12.2			13.7	
THETA_6 (L/h/70kg)	11.4			8.86	

Model scraping for comparison and validation

Between-subject variability (inter-individual variability)					
OMEGA_CL (CV %)			47.1	35	
OMEGA_V1 (CV %)			39.2	31.9	
OMEGA_V2 (CV %)			47.5	43.8	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.228			20.1	
Notes/further info:					

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	NA
Notes/further info:			